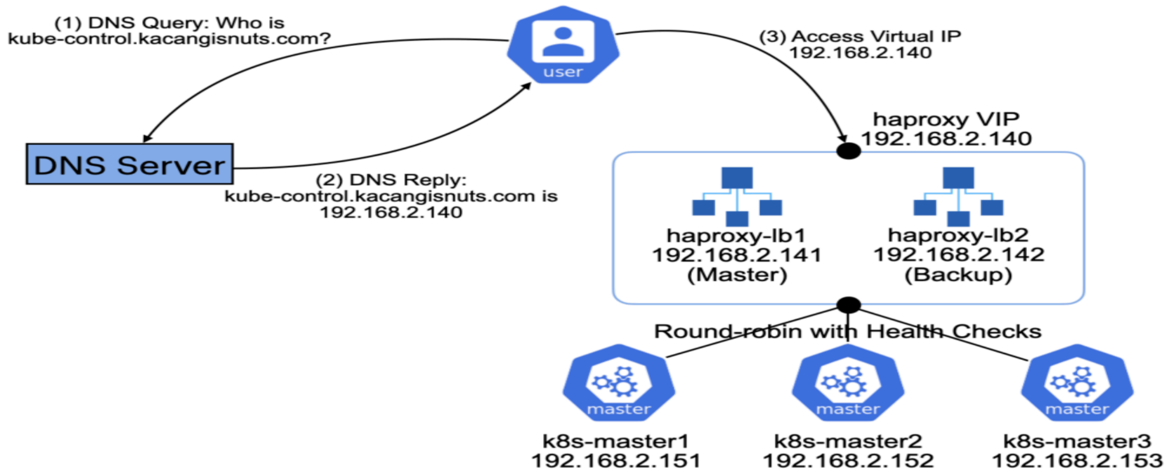
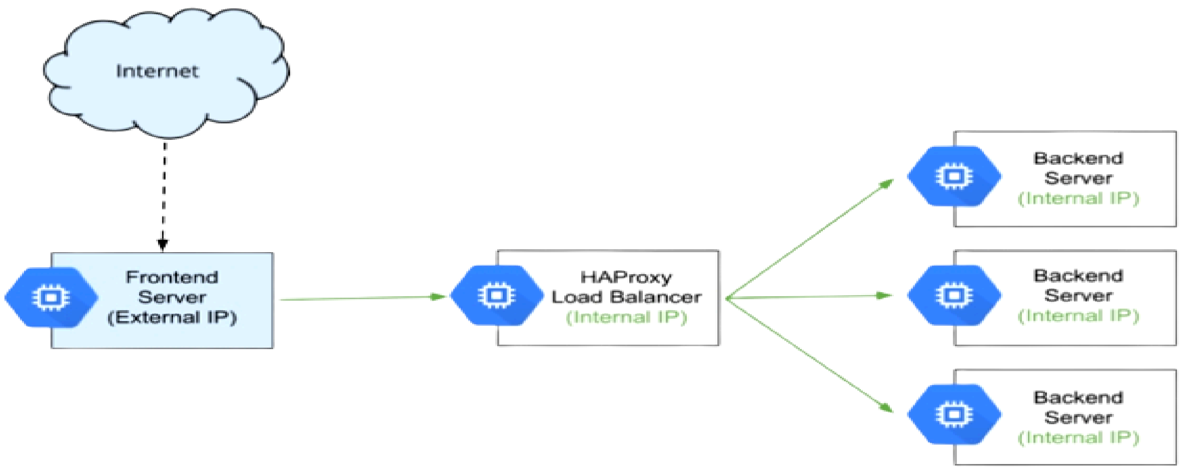
Setup HAProxy

HAProxy is a free, very fast and reliable reverse-proxy offering high availability, load balancing, and proxying for TCP and HTTP-based applications. It is particularly suited for very high traffic web sites and powers a significant portion of the world's most visited ones. Over the years it has become the de-facto standard open source load balancer, is now shipped with most mainstream Linux distributions, and is often deployed by default in cloud platforms.





Tasks:

1. Hardware Recommendations
2. Installing HAproxy load balancer
3. Configuring HAproxy as a load balancer
4. Configuring HAproxy Monitoring
5. Managing the HAproxy server
6. Configure hosts file
7. Test HAProxy using a web browser
8. Testing HAproxy Monitoring

STEP 1 :

Hardware Recommendations :

The hardware requirements for HAProxy Enterprise depend on the workload it needs to manage:

- Only CPU and memory are taken into consideration.
- Disk size depends on your operating system and the volume of logs you want to keep.

Workload	Tasks	Sizing
Low-level	• TCP or HTTP traffic • Up to 1000 conn/s • Very low SSL traffic or gzip compression	This type of workload can be achieved either by a Virtual Machine or a bare metal server. You need at least: • 1 CPU core • 1 G of RAM
Mid-level	• TCP or HTTP traffic (including HTTP manipulation) • Up to 4000 conn/s • Low SSL traffic or gzip compression	This type of workload can be achieved either by a Virtual Machine or a bare metal server. You need at least: • 2 CPU cores • 1 G of RAM
High-level	• TCP or HTTP traffic (including HTTP manipulation) • Up to 20000 conn/s • 10% of traffic ciphered (SSL) or compressed	This type of workload can be achieved by a bare metal server only. You need at least: • 2 CPU cores, as fast as possible • 4G of RAM • powerful network card

STEP 2 :

Installing HAproxy load balancer :

Then update packages using the below command:

sudo apt-get update

sudo apt-get upgrade

Now install HAproxy using the following command in Terminal :

sudo apt install haproxy

Once the installation of the HAproxy server is finished, you can confirm it using the below command in Terminal :

haproxy -v

It will show you the installed version of HAproxy on your system which verifies that the HAproxy has been successfully installed.

STEP 3 :

Configuring HAproxy as a load balancer :

we will configure HAproxy as a load balancer. To do so, edit the **/etc/haproxy/haproxy.cfg** file :

sudo nano /etc/haproxy/haproxy.cfg

```
frontend http_front
  bind x.x.x.x:80
  stats uri /haproxy?stats
  default_backend x
```

backend x

balance roundrobin

server worker1 x.x.x.x:x weight 1 check

server worker2 x.x.x.x:x weight 1 check

STEP 4 :

Configuring HAproxy Monitoring :

With HAproxy monitoring, you can view a lot of information including server status, data transferred, uptime, session rate, etc. To configure HAproxy monitoring, append the following lines in the configuration file located at **/etc/haproxy/haproxy.cfg**:

sudo nano /etc/haproxy/haproxy.cfg

listen stats

bind x.x.x.x:x

mode http

option forwardfor

option httpclose

stats enable

stats show-legends

stats refresh 5s

stats uri /stats

stats realm Haproxy\ Statistics

stats auth x:x   #Login User and Password for the monitoring

stats admin if TRUE

default_backend x

Once you are done with the configurations, save and close the **haproxy.cfg file**.

Now verify the configuration file using the below command in Terminal:

haproxy -c -f /etc/haproxy/haproxy.cfg

Now to apply the configurations, restart the HAproxy service:

sudo systemctl restart haproxy.service

If get error change port config : port 80 *

STEP 5 :

Managing the HAproxy server: :

Here are some other commands for managing the HAproxy server:

In order to start the HAproxy server, the command would be:

sudo systemctl start haproxy.service

In order to stop the HAproxy server, the command would be:

sudo systemctl stop haproxy.service

In case you want to temporarily disable the HAproxy server, the command would be:

sudo systemctl disable haproxy.service

To re-enable the HAproxy server, the command would be :

sudo systemctl enable haproxy.service

STEP 6 :

Configure hosts file :

Edit the /etc/hosts file using the below command in Terminal:

sudo nano /etc/hosts

Add the following hostname entries for both Kubernetes workers along with its own hostname:

x.17.21.108 x

x.17.31.75   x

x.17.31.28   x

Now save and close the /etc/hosts file.

STEP 7 :

Test HAProxy using a web browser :

Now in your HAproxy server, open any web browser and type :

http://x.com

STEP 8 :

Testing HAproxy Monitoring :

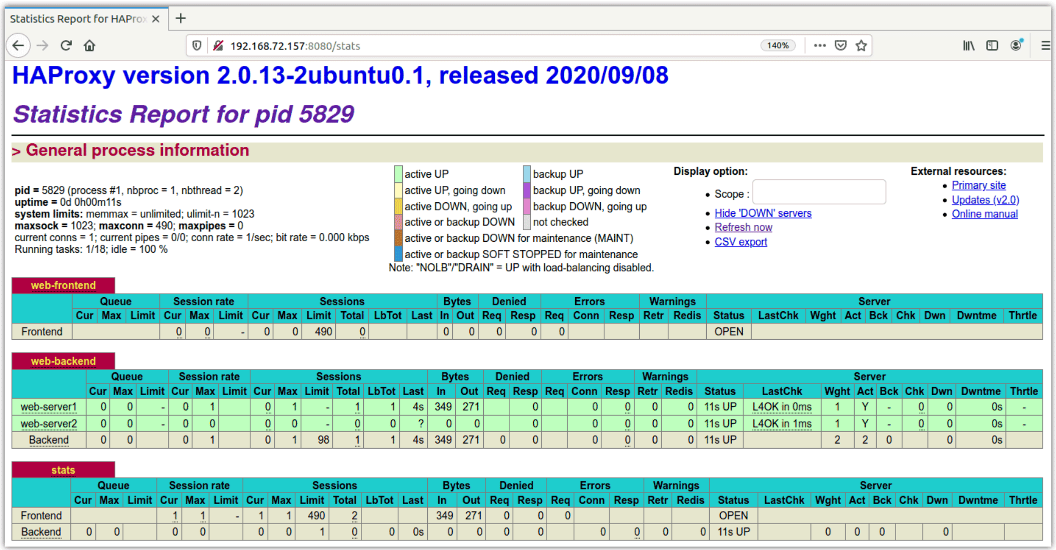
To access the HAproxy monitoring page, type http:// followed by the IP address/hostname of HAproxy server and port 8080/stats:

http://haproxy-ip:8080/stats

username : x

password : x

This is the statistics report for our HAproxy server



Statistics Report for HAProxy

192.168.72.157:8080/stats

HAProxy version 2.0.13-2ubuntu0.1, released 2020/09/08

Statistics Report for pid 5829

> General process information

pid = 5829 (process #1, nbproc = 1, nbthread = 2)
uptime = 0d 0h00m11s
system limits: memmax = unlimited; ulimit-n = 1023
maxsock = 1023; **maxconn =** 490; **maxpipes =** 0
current conns = 1; current pipes = 0/0; conn rate = 1/sec; bit rate = 0.000 kbps
Running tasks: 1/18; idle = 100 %

active UP
backup UP
active UP, going down
backup UP, going down
active DOWN, going up
backup DOWN, going up
active or backup DOWN
not checked
active or backup DOWN for maintenance (MAINT)
active or backup SOFT STOPPED for maintenance
Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

Display option:
- Scope :
- Hide 'DOWN' servers
- Refresh now
- CSV export

External resources:
- Primary site
- Updates (v2.0)
- Online manual

web-frontend

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server							
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle
Frontend				0	0	-	0	0	490	0			0	0	0	0	0					OPEN								

web-backend

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server							
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle
web-server1	0	0	-	0	1		0	1	-	1	1	4s	349	271		0		0	0	0	0	11s UP	L4OK in 0ms	1	Y	-	0	0	0s	-
web-server2	0	0	-	0	0		0	0	-	0	0	?	0	0		0		0	0	0	0	11s UP	L4OK in 1ms	1	Y	-	0	0	0s	-
Backend	0	0		0	1		0	1	98	1	1	4s	349	271	0	0		0	0	0	0	11s UP		2	2	0		0	0s	

stats

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server							
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle
Frontend				1	1	-	1	1	490	2			349	271	0	0	0					OPEN								
Backend	0	0		0	0		0	0	1	0	0	0s	0	0	0	0		0	0	0	0	11s UP		0	0	0		0		

Successful

END